SHREE VARAA MANGAI V

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SKILLS

ML / LLM & AI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, LangChain, LlamaIndex, RAG, Prompt Engineering, OpenAI API, Gemini API, Pinecone, Gradio, Streamlit, CUDA, Weights & Biases (wandb), Matplotlib

Full-Stack Engineering: Python, TypeScript, JavaScript, Java, C/C++, Dart · React.js, Node.js, Flask, FastAPI · PostgreSQL, MongoDB, Firebase, SQL, REST APIs, Swagger

Deployment & Tools: Docker, Git, Linux, CI/CD Pipelines, Azure, Postman

Data & Visualisation: PowerBI, Tableau, Matlab, Chart.js, Microsoft Office, Google Workspace, Canva

Soft Skills: Team Management, Agile Collaboration, Communication, Mentoring, Problem Solving, Time Management

WORK EXPERIENCE

ML Engineering Co-op | University of Florida- IPPD | AGIS Inc. *Florida, US | Aug 2025 - Apr 2026*

It's an enterprise AI company that builds custom AI-powered tools for business automation. This co-op focused on building a real-time 3D asset reconstruction pipeline from video/image streams for Unity-based visualization.

- Built an end-to-end pipeline that captures video/ image streams, reconstructs them into 3D assets using 2D Gaussian Splatting (2DGS), and integrates them into Unity for real-time interactive visualization.
- Designed and trained neural reconstruction models (3DGS/2DGS, COLMAP/GLOMAP) on multi-view data to generate high-fidelity meshes and point clouds; selected 2DGS for its superior exportable mesh quality over 3DGS.
- Accelerated large-scale reconstruction and rendering using NVIDIA B200 GPU clusters with CUDA nightly builds, significantly improving training throughput and pipeline scalability.
- Investigated real-time vs. offline processing tradeoffs to optimize computational efficiency, reducing reconstruction latency and enabling practical deployment workflows.

Machine Learning Researcher | University of Florida- Trustworthy Engineered Autonomy Lab *Florida, US | Sep 2025 - Apr 2026*

Research lab focused on reliable and interpretable AI systems. This role centred on benchmarking hallucination in text-to-video generative models.

- Designed and executed experiments generating videos with Wan 2.1 T2V 1.3B and HunyuanVideo from T2VCompBench and ViBe benchmark prompts, systematically quantifying semantic discrepancies between prompts and generated content.
- Developed a fine-grained hallucination taxonomy identifying object omissions, attribute mismatches, spatial relationship errors, and semantic drift across video generation stages.
- Benchmarked automated hallucination detection using Qwen3-VL and other VLMs on severity-level classification; evaluated using Balanced Accuracy, Macro F1, AUROC, and AUPRC.

Global People Analytics Intern | Ford Motor Company *Tamil Nadu, IN | Aug 2023 - Oct 2023*

Ford's Global People Analytics team uses data-driven methods to inform HR strategy. This internship focused on workforce forecasting and budget planning.

- Conducted statistical analysis and forecasting on time series forecasting models such as Autoregressive Integrated Moving Average (ARIMA), Vector Autoregression (VAR), and Vector Error Correction Model (VECM) and non-time series models, such as Lasso and Ridge regression, to determine optimal female incumbency.
- Developed ETL pipelines in Python to predict salary costs using workforce datasets containing 1,000+ employee records, improving model accuracy by 20% through exploratory data analysis, feature engineering, and data quality improvements.

Software Engineer Intern | Spacescan Ltd. *Ontario, Canada (Remote) | Sep 2022 - Jan 2023*

Spacescan is a blockchain explorer and analytics platform. This internship involved full-stack development of the consumer-facing web and mobile application.

- Built and maintained responsive web interfaces in React.js, integrating REST APIs documented with Swagger to enable seamless frontend-backend communication over PostgreSQL databases.
- Developed reusable, scalable UI components with robust state management patterns, reducing code duplication and improving long-term maintainability of the application.
- Contributed to the React Native mobile app, implementing new features and participating in frontend architecture discussions to ensure consistency across web and mobile platforms.

PROJECTS

CareMind: Agentic Clinical RAG Assistant | *GitHub-Link* *May - Jun 2026*

- Built an agentic RAG system using LangGraph for multi-step query routing across document retrieval, report comparison, and medical education workflows over synthetic medical PDFs.
- Engineered an MCP tool server for document search, report comparison, and timeline extraction; integrated tool-call results into cited answers using NVIDIA NIM embeddings and Pinecone vector search.
- Deployed a React frontend with a FastAPI backend, Redis response cache, and a repeatable evaluation pipeline measuring route accuracy and citation pass rate; accessible via both web and VS Code extension.

Tech Stack: Python, FastAPI, LangGraph, Pinecone, Redis, NVIDIA NIM, MCP, React, TypeScript

Multi-Image Super Resolution for Prostate MRI (CNN, SRGAN, Diffusion Models) | *GitHub-Link* *Nov - Dec 2025*

- Developed a multi-image super-resolution pipeline to reconstruct missing or low-resolution prostate MRI slices, targeting clinical-grade anatomical consistency.
- Evaluated CNN, SRGAN, and diffusion-based architectures on PSNR, SSIM, and qualitative anatomical assessment, analysing tradeoffs between perceptual sharpness and structural fidelity to minimise hallucinated structures.

Tech Stack: PyTorch, CNN, GAN, Diffusion Models, OpenCV, NumPy

Multimodal Spoken Command Recognition (Audio-Text Fusion + Vector Retrieval) | *GitHub-Link* *Mar - Apr 2025*

- Developed a multimodal classification model fusing Wav2Vec2 audio embeddings and BERT semantic embeddings via cross-attention and Transformer decoders to classify spoken commands.
- Integrated Pinecone vector retrieval to semantically enhance inference; explored fusion strategies and documented generalization challenges under noisy conditions to guide future robustness improvements.

Tech Stack: PyTorch, Hugging Face Transformers, Wav2Vec2, BERT, Pinecone, Scikit-learn

CognitoMap: Automated Question Classification using Bloom's Taxonomy | *GitHub-Link* *Feb - Apr 2024*

- Built a scalable ETL pipeline extracting and transforming questions from PDFs into MongoDB, achieving 94% accuracy in question extraction using the Gemini 1.0 Pro model; classified questions into Bloom's Taxonomy levels using Sentence Transformer embeddings with Pinecone vector retrieval.
- Delivered a full-stack dashboard (Flask, React.js, Chart.js) giving educators real-time insights into the cognitive distribution of their assessments across Bloom's Taxonomy levels.

Tech Stack: Flask, React, MongoDB, Pinecone, Chart.js, TypeScript

CERTIFICATIONS

- Building with the Claude API - Anthropic (Jun 2026)
- Build and Orchestrate Agents with Microsoft Foundry - Microsoft AI Skills Challenge (Jun 2026)
- Fundamentals of Deep Learning - NVIDIA (Nov 2024)
- Fine-Tuning LLMs for Cybersecurity (Mistral, Llama, AutoTrain, AutoGen, LLM Agents) - LinkedIn Learning (Jan 2025)
- Tesla Stock Price Prediction & Google Cloud Fundamentals - Coursera (Dec 2023 / Aug 2022)
- Cloud Computing & Distributed Systems; Java Programming - NPTEL (Mar 2021 / Oct 2021)

EDUCATION

Master of Science in Artificial Intelligence Systems | GPA: 3.90 / 4.0 *Aug 2024 - May 2026*

University of Florida, Gainesville, FL

Relevant Coursework: Deep Learning in Medical Image Analysis, ML for AI Systems, Applied Deep Learning, AI Ethics, Computational Photography & AI Systems Project

Bachelor of Technology in Information Technology | GPA: 8.43 / 10 *Sep 2020 - May 2024*

Loyola-ICAM College of Engineering and Technology, India

Relevant Coursework: OOP Principles, Computational Intelligence, Cryptography & Network Security, Design and Analysis of Algorithms, Big Data Analytics, Linear Algebra, Statistics, Database Management Systems, Data Structures